



## DAMAGE ANALYSIS OF ALUMINUM MATRIX COMPOSITE CONSIDERING NON-UNIFORM DISTRIBUTION OF SiC PARTICLES

M. GENI<sup>1</sup> and M. KIKUCHI<sup>2</sup>

<sup>1</sup>Xinjiang Engineering University, Urumqi, Xinjiang 830008, China and <sup>2</sup>Department of Mech. Engng Sciences, University of Tokyo, Noda, Chiba, Yamasaki 2641, Japan

(Received 26 June 1997; accepted in revised form 11 December 1997)

**Abstract**—In this study, based on the fracture surface observations using an SEM (scanning electron microscope), a numerical unit model and 3-dimensional non-uniform model for FEM (finite element method) analysis are proposed by using Gurson's model as a constitutive equation. The effect of the non-uniform distribution of the SiC particle volume fraction and aspect ratio in the matrix are considered for the evaluation of the stress-strain relationship and the damage. The results show that the dimple fracture process of matrix aluminum alloy is well simulated, after a large amount of plastic deformation. The non-uniformity of the SiC particle volume fraction, aspect ratio and its locations have a strong effect on the local and global damage behavior and stress-strain relationships. It is shown that the fracture strain increases largely when the aspect ratio of SiC particles is 1.0 and it is distributed uniformly in the matrix. The numerical model is improved by considering the non-uniform effect on the tensile stress, fracture strain and local and global fracture. © 1998 Acta Metallurgica Inc.

### 1. INTRODUCTION

SiC particle reinforced aluminum matrix composites achieve high yield stress, high strength and high stiffness by mixing stiff reinforcement SiC particles into the aluminum alloy. The density of this composite material is decreased by increasing the volume of SiC particle. For these advantages, this material is attracting attention for potential applications. However, the use of this material is limited by its low fracture toughness and ductility. In this material the particle shape, aspect ratio and their distribution are not uniform in the matrix. This non-uniformity affects the local stress-strain state and the local constraint condition on the ductile behaviour of the materials. It is important to study the effect of these non-uniform factors on the fracture strength of the material. For this purpose, both experimental and computational studies have been conducted by many authors [1–10].

The experimental analyses have been conducted by Kamat *et al.* [11], Flom *et al.* [12], Tao *et al.* [6] and Kikuchi *et al.* [13]. These authors showed that the damage of this material is due to the dimple fracture of the matrix, particle cracking and the debonding between a particle and the matrix. Kamat *et al.* indicated that the yield strength increases with decreasing particle spacing. Flom *et al.* concluded that the fracture behavior of this material mainly depends on the particle volume fraction. Tao *et al.* showed that the fracture toughness

of this material decreases with the increase of the standard deviation of local particle volume fraction and nearest particle spacing. Kikuchi *et al.* showed that the volume fraction of SiC particles on the fracture surface is much larger than the average value. Apparently the fracture occurs where the SiC particle volume fraction is comparatively larger than in other areas and the final fracture occurs as a result of the linkage of these locally fractured areas.

The numerical damage analyses of SiC whisker or SiC particles reinforced aluminum alloys have been conducted by Christman *et al.* [14], Kikuchi *et al.* [13,15], Finot *et al.* [16] and Shen *et al.* [17] using the unit cell models. Christman *et al.* conducted unit cell analysis by considering the effect of thermal residual stress and Kikuchi *et al.* carried out unit cell simulations by considering the void nucleation and growth effect in the base matrix materials. Finot *et al.* reported the effects of particle size, local clustering of particles, particle volume fraction and mode of applied loading on propensity for particle fracture in the matrix. Shen *et al.* discussed the effects of particle shape and thermal residual stress. These numerical results show that the stress-strain relationships agree with the experimental ones qualitatively and the local damage processes are well simulated. But the effect of non-uniform distribution of whiskers or particles is not addressed explicitly. One reason for this is that the

cell models are made by assuming that the matrix contains a periodic array of whiskers or particles with identical size, shape and distance or with identical distribution of particles. In the MMC (metal matrix composite) materials, the non-uniform distribution of local particle volume fraction, particle spacing and size greatly affects the damage behavior. In addition, the dependence of the damage process on particle size and aspect ratios, mean and local particle volume fraction and the nearest particle spacing may vary in different local areas, but this non-uniform effect has not been thoroughly studied yet.

In this study, based on the basic tensile tests and fracture surface observations by SEM, axisymmetric cell models and 3-dimensional non-uniform models for FEM analyses are proposed. Numerical studies have been conducted using the damage mechanics approach for the fracture process of SiC particle reinforced aluminum alloys. The effects of the non-uniform distribution of SiC particle volume fraction and the aspect ratio in the matrix are studied.

## 2. SUMMARY OF TENSILE TESTS

Table 1 shows the mechanical properties of SiC particulate-reinforced aluminum matrix composites, base matrix and SiC particle. The 0.2% proof stress  $\sigma_{0.2}$  and the ultimate strength  $\sigma_U$  are measured experimentally by the authors with the exception of the SiC particle. The elastic constants are measured by the ultrasonic pulse method [18]. Mechanical properties of the SiC particle are taken from Refs [14, 19]. The base matrix is 2024-T6 aluminum alloy. Two kinds of SiC particulate reinforced materials are made through powder metallurgy by changing the volume fraction of SiC in two cases, 2 and 10%. They are called SiC2% and SiC10% specimens, respectively. A specimen of the base matrix, called SiC0% specimen, is also tested. The average diameter of SiC particles is 4  $\mu\text{m}$ . The tensile test is conducted keeping the cross head speed at 0.5 mm/min.

Figure 1 is the SEM photograph of the fracture surface after the test. The dimples and SiC particles are observed in this photograph. In the authors' previous paper [15], the dimple pattern of matrix, particle cracking and the debonding between a particle and the matrix are observed using the SEM matching photos. The mean volume fractions of

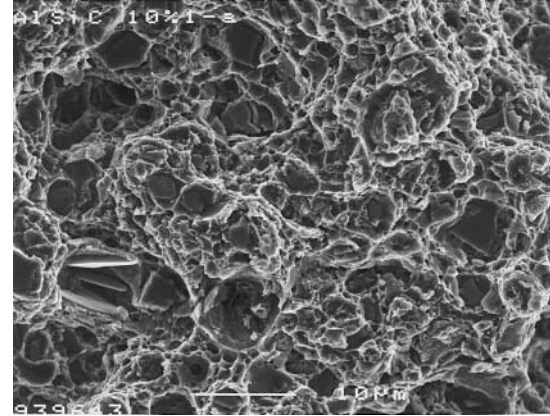


Fig. 1. Dimple fracture of SiCp/2024-T6 specimen (SiC10%).

SiC particles on the fracture surfaces are 17% for SiC10% specimen and 11% for SiC2% specimen, which are much larger than the average value.

Figure 2 shows the frequency distribution of SiC particle aspect ratio observed on the fracture surface. The aspect ratio,  $A_p$ , is defined as the ratio of the long diameters,  $b$ , to the short ones (black columns in Fig. 2) and short diameters divided by long ones (white columns in Fig. 2). Here only two directions, the long diameter of SiC particles parallel to the loading direction and the one perpendicular to it, are considered. It is shown that the frequencies of aspect ratios 0.5 and 2.0 are much larger than those of the aspect ratios 0.25 and 4.0. The average aspect ratio  $A_p$  of the whole particles are  $a/b = 0.63$  and  $b/a = 1.77$ .

The fracture of SiC particulate-reinforced aluminum alloy is classified into three microscopic processes. They are: (1) dimple fracture of matrix, (2)

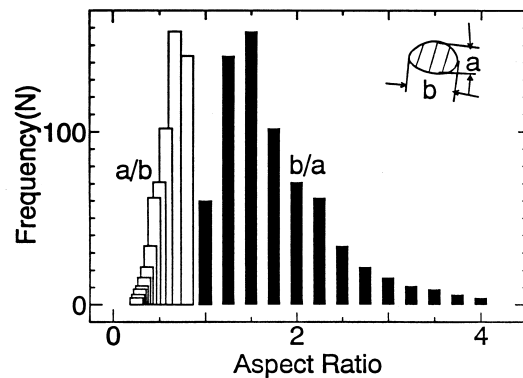


Fig. 2. Distribution of particles aspect ratio.

Table 1. Mechanical properties

| Materials             | $\sigma_{0.2}$ (MPa) | $\sigma_B$ (MPa) | $E$ (GPa)                                                | $\nu$ |
|-----------------------|----------------------|------------------|----------------------------------------------------------|-------|
| Al 2024-T6            | 383.2                | 570              | 75.47                                                    | 0.333 |
| SiCp/2024-T6 2% vol.  | 445                  | 547              | $\sigma = 897.4(0.00538 + \epsilon^p)^{0.1628}$<br>77.35 | 0.330 |
| SiCp/2024-T6 10% vol. | 450                  | 575              | 88.03                                                    | 0.325 |
| SiCp [14, 19]         | 2450                 | —                | 450                                                      | 0.17  |

Table 2. Values of  $f_N$ ,  $f_C$  and  $f_F$ 

| Materials                           | Parameter          |                    |       |
|-------------------------------------|--------------------|--------------------|-------|
|                                     | $f_N$              | $f_C$              | $f_F$ |
| Rigid-perfectly plastic matrix [28] | 0.04               | 0.15               | 0.25  |
| Al 2017-T4 [26]                     | 0.013              | 0.169              | 0.206 |
| SiCp/2024-T6 for 2 and 10% vol.     | 0.013 <sup>a</sup> | 0.085 <sup>b</sup> | 0.142 |

<sup>a</sup>Assumed.<sup>b</sup>Estimated.

debonding of matrix and particle and (3) particle cracking. The non-uniform distribution of particles, especially in the local area where the SiC volume fraction is large, affects these phenomena strongly. The same experimental results are shown in Refs [6, 11, 20].

### 3. NUMERICAL MODEL

Due to the complexity, it was not possible to determine an exact measurement of interface mechanical properties. In the following analysis, the perfectly bonded interface and rigid particles are assumed. The dimple fracture of the matrix is considered in the local damage process. The non-uniform distribution of particles volume fraction and their aspect ratios are also considered in the global damage process.

#### 3.1. The model of matrix dimple fracture

Though the damage in this material occurs by different mechanisms, the base matrix aluminum alloy mainly shows dimple fracture which occurs due to the nucleation and growth of micro voids in the base matrix. Early investigations studied the void evolution in ductile material. McClintock [21] studied the evolution of cylindrical voids and Rice and Tracey [22] considered the spherical voids in infinite rigid perfectly plastic media. Gurson [23] developed an approximate model for ductile metals containing cylindrical and spherical voids. Gurson assumed that the void is distributed randomly in the matrix, so that the global response of the model is isotropic. Then Gurson proposed the yield condition for a spherical model containing a void volume fraction,  $f$ , as follows:

$$\Phi = \frac{\sigma_e^2}{\sigma_m^2} + 2q_1f \cosh\left\{\frac{q_2\sigma_{kk}}{2\sigma_m}\right\} - 1 - (q_1f)^2 = 0 \quad (1)$$

where  $\sigma_e$  is the equivalent stress,  $f$  is the void volume fraction and  $\sigma_m$  is the equivalent stress of the matrix. The constants  $q_1$  and  $q_2$  are introduced

by Tvergaard [24]. The change of the value  $f$  is modified by Tvergaard and Needleman [25] to model the loss of stress carrying capacity of the matrix as follows:

$$f = \begin{cases} f, & \text{for } f \leq f_C \\ f_C + \frac{1/q_1 - f_C}{f_F - f_C}(f - f_C), & \text{for } f > f_C \end{cases} \quad (2)$$

where  $f_C$  is the critical void volume fraction and  $f_F$  is the value of  $f$  at fracture. Table 2 lists the values of these parameters used in the numerical analyses.  $f_F$  is decided by the average dimple diameter as shown in Table 3. It is shown that the average dimple diameter of the SiC0% base matrix aluminum alloy is nearly 8 times larger than those of SiC2% and SiC10% specimens. It is likely due to the fact that the plastic flow of the matrix is constrained by the reinforcing SiC particles.  $f_N$  is the volume fraction of segregated particles. Its value is assumed to be equal to that of 2017-T4 aluminum alloy, which is measured by Okayasu *et al.* [26]. The  $f_C$  value is estimated from the ratio of  $f_F$  to  $f_C$  for rigid-perfectly plastic matrix material which are chosen based on the reasonable experimental results by Brown and Embury [27] and with numerical results by Anderson [28].

In the homogeneous ductile materials this modified Gurson model has been compared with other experiments and consistency was shown by many authors [29]. Previous papers [3, 13] discussed the tensile properties of SiC whisker and particle reinforced aluminum alloy using a modified Gurson's model. The results show that the numerical simulation agrees well with the experimental one qualitatively. In this analysis the same method is used.

#### 3.2. The model for non-uniformly distributed particles

From the global viewpoint, it seems that the SiC particles are distributed uniformly given a sufficient area of the material, but the damage does not uniformly occur in the materials. Due to the shape of the materials and non-uniform distribution of SiC

Table 3. Dimple diameter

| Materials              | Parameter      |                                   |                             |
|------------------------|----------------|-----------------------------------|-----------------------------|
|                        | No. of dimples | dimple diameter ( $\mu\text{m}$ ) | void volume fraction, $f_F$ |
| Al 2024-T6             | 260            | 4.7                               | 0.26                        |
| SiCp/2024-T6, 2% vol.  | 280            | 0.56                              | 0.142                       |
| SiCp/2024-T6, 10% vol. | 460            | 0.56                              | 0.142                       |

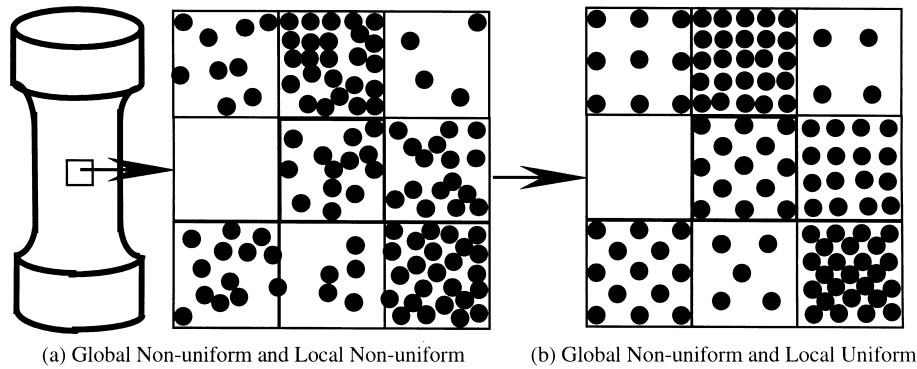


Fig. 3. The non-uniform model. (a) Global non-uniform and local non-uniform and (b) global non-uniform and local uniform.

particles, the damage occurs in some local areas where the stress concentration is large. These defects (damaged areas or stress concentrated areas) cause the damaged areas to expand. Due to the increasing elastic plastic deformation, the damage concentrations occur at some local areas and its non-uniformity becomes strong in the materials. This indicates that the complicated shape and non-uniform distribution of particles have large effects on the damage process.

To consider the effect of the particles non-uniform distribution on damage process, non-uniform numerical models have to be developed. Figure 3(a) and (b) show the scheme of a non-uniform model. In the real structure, as shown in Fig. 3(a), the SiC reinforcements are distributed non-uniformly locally and globally. Due to the limit of the computer memory it is difficult to model all of the SiC reinforcement in the matrix. Here a very simple model is considered and a convenient approximation is conducted for the modelling of the non-uniform distribution of SiC particles. It assumes that the SiC particles are distributed uniformly locally and non-uniformly globally as shown in Fig. 3(b). Thus, different local cell models are obtained easily by considering the symmetry of the local structure. In Section 4 this cell model is called a unit model. The non-uniform models can be made by composing several unit models with different SiC particle volume fractions, aspect ratios and shapes.

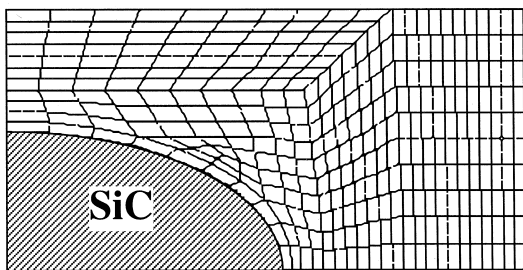


Fig. 4. The mesh pattern of unit model ( $Ap = 0.5$ ).

Figure 4 shows the FEM mesh pattern of a axisymmetric unit models with the SiC particle volume fraction of 10% and aspect ratio is 0.5. Aspect ratios of 0.25, 1.0, 2.0 and 4.0 models were also made. These models are called  $Ap = 0.25$ ,  $Ap = 0.5$ ,  $Ap = 1.0$ ,  $Ap = 2.0$  and  $Ap = 4.0$  models with SiC10% volume fraction, respectively, in the following analysis. Five unit models were made by changing the SiC volume fractions to 0, 2, 10, 21 and 31% and by keeping the aspect ratios as 1.0 and 4.0, respectively. These are called SiC0%, SiC2%, SiC10%, SiC21% and SiC31% models with  $Ap = 1.0$  and  $Ap = 4.0$ . The shapes of the SiC particles are ellipses of revolution as oblate spheroid (an aspect ratio less than 1.0), spherical (an aspect ratio equal to 1.0) and prolate spheroid (an aspect ratio greater than 1.0). The unit models are made by keeping the unit model aspect ratio equal to the SiC particle aspect ratio and keeping the mesh sizes nearly equal.

#### 4. RESULTS

##### 4.1. The effect of particle aspect ratio

Figure 5 shows the distribution of void volume fraction,  $f$ , around the oblate ( $Ap = 0.5$ ), spherical

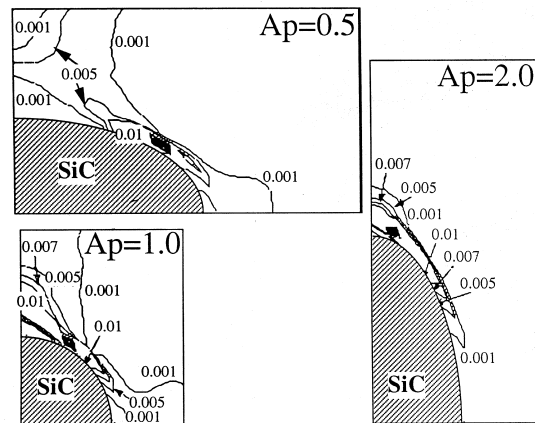


Fig. 5. The distributions of void volume fraction.

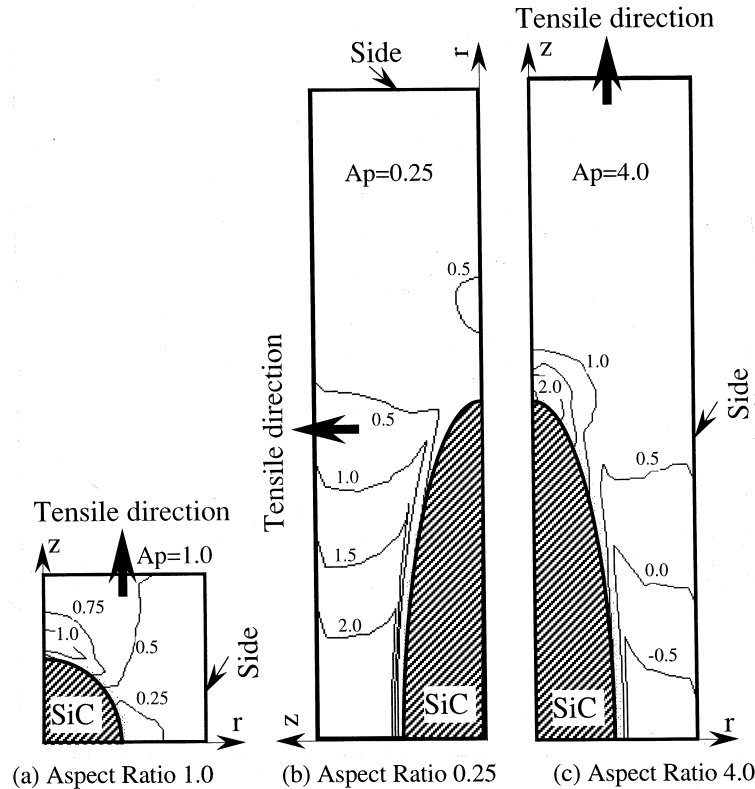


Fig. 6. The distributions of stress triaxiality. (a) Aspect ratio, 1.0, (b) aspect ratio, 0.25 and (c) aspect ratio, 4.0.

( $Ap = 1.0$ ) and prolate ( $Ap = 2.0$ ) particle models just after the first dimple fracture occurs. It is shown that the void volume fraction distributions are changed by changing the aspect ratio of the SiC particle. In the spherical particle, the largest  $f$  value area appears near the particle/matrix interface. The void volume fractions contour lines  $f = 0.001-0.01$  do not grow along the particle/matrix interface and spread into the matrix. In the oblate particle, the void is nucleated on the particle surface and grows along the particle/matrix interface. Some contour lines around  $f = 0.001-0.005$  also spread into the matrix. In the prolate particle, the void volume fraction contour lines expressed by  $f = 0.001-0.01$  are rather concentrated along the particle/matrix interface. It is shown that the first dimple fracture of the matrix occurs at the particle/matrix interface boundary and the growing configuration of the dimple fracture depends on the particle aspect ratio.

Figure 6(a), (b) and (c) show the distribution of stress triaxiality ( $\sigma_{kk}/(3\sigma_e)$ ) of spherical ( $Ap = 1.0$ ), oblate ( $Ap = 0.25$ ) and prolate ( $Ap = 4.0$ ) models when the global strain is 0.015. It is shown that the stress triaxiality contour lines are changed in the matrix roughly from 0.25 to 1.0 in the  $Ap = 1.0$  model, from 0.5 to 2.0 in the  $Ap = 0.25$  model and from  $-0.5$  to 2.0 in the  $Ap = 4.0$  model. Generally, it is considered that the side constraint becomes stronger due to the non-uniform distribution of side

stress triaxiality along the side. This means that the side constraint becomes stronger by increasing the particle aspect ratio.

Figure 7 shows the global stress-strain relationships of  $Ap = 0.25$ ,  $Ap = 0.5$ ,  $Ap = 1.0$ ,  $Ap = 2.0$  and  $Ap = 4.0$  models with SiC10% volume fraction, respectively. As the stress decreases suddenly just after the fracture of the first element, critical fracture strain and fracture stress are defined at this position. The results show that the ultimate strength increases by increasing the SiC particle aspect ratio.

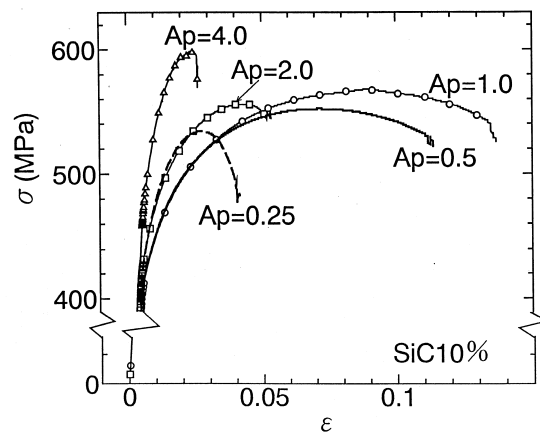


Fig. 7. The stress-strain relationships (the effect of aspect ratio,  $Ap$ ).

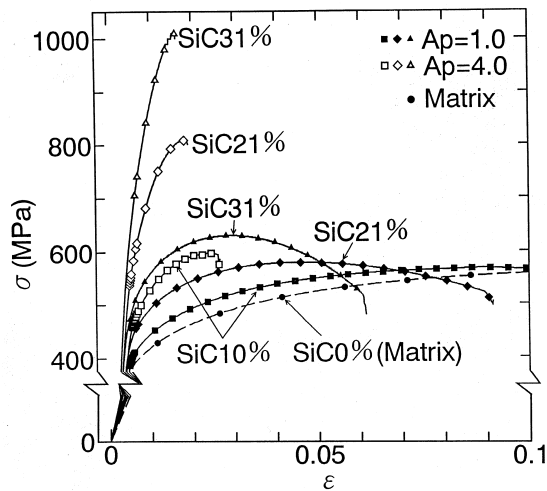


Fig. 8. The stress-strain relationships (the effect of SiC volume fraction).

This is due to the fact that the side constraint becomes stronger and the strain constrained areas are increased in the matrix by increasing the SiC particle aspect ratio as shown in Fig. 6. It is shown that the fracture strain increases with the aspect ratio and it decreases when the aspect ratio exceeds 1.0. The reason for this is that the distribution of void volume fraction around the SiC particle interface is changed by changing the aspect ratio of the SiC particle as shown in Fig. 5. If the aspect ratio

is equal to 1.0, the void volume fraction spreads into the matrix and disperses into the area losing a wide range of its stress carrying capacity. Then it improves the load carrying capacity in these areas and the fracture strain increases. But if the aspect ratio is less than 1.0 or greater than 1.0, the void grows along the particle/matrix interface and the area losing the stress carrying capacity is restricted within narrow area. Then the fracture strain is decreased. It is obvious that the fracture strain is increased by increasing the uniform distribution of void nucleation, growth and coalescence in the matrix. As shown in Fig. 7, the aspect ratio equal to 1.0 model gives the largest fracture strain and the ultimate strength nearly equal to that of the model where the aspect ratio is equal to 2.0.

Figure 8 shows the stress-strain relationship of five unit models with different SiC particle volume fraction and with aspect ratios 1.0 and 4.0. When the aspect ratio increases from 1.0 to 4.0 in the SiC31% models, the ultimate strength increases about 400 MPa. In SiC10% models, it increases approximately 50 MPa. Due to the decrease of the SiC particle volume fraction, the effect of the aspect ratio on the ultimate strength becomes weak. This is due to the fact that the deformation constraint between SiC particles becomes weak by decreasing the particle volume fraction and the deformation constraint by the aspect ratio also becomes weak. One interesting result in Fig. 8 is that the ultimate

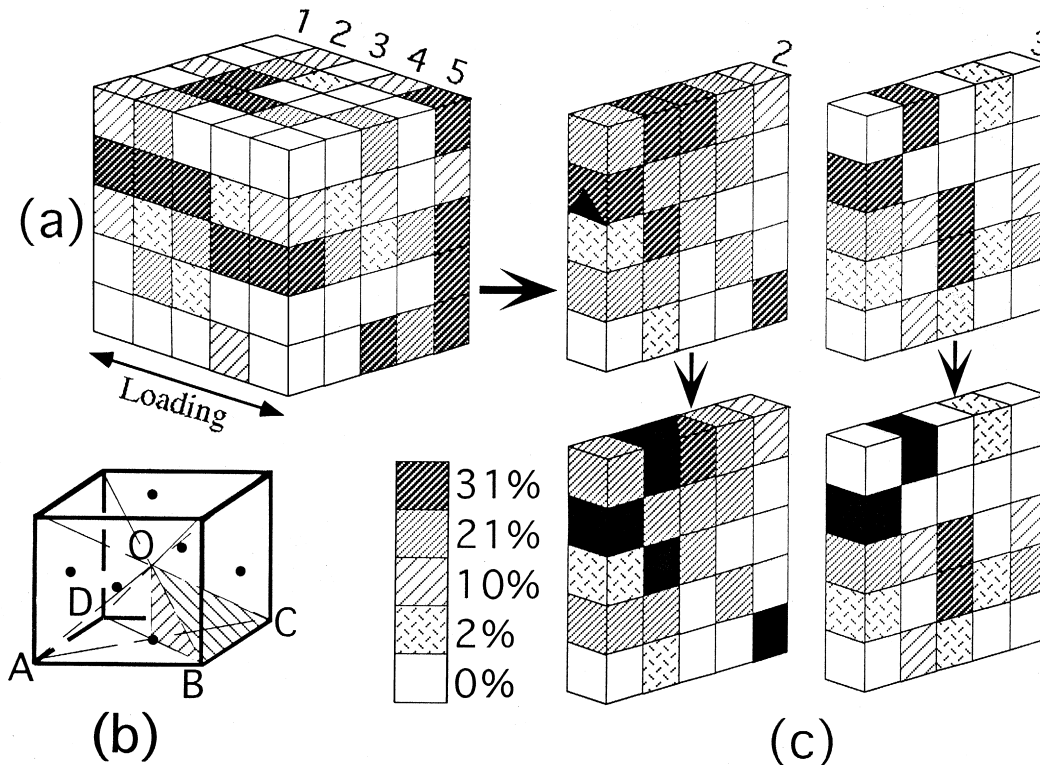


Fig. 9. The fracture process in RV10A1 non-uniform model (the effect of particle volume fraction). (a) Non-uniform model, (b) tetrahedral element and (c) fracture process.

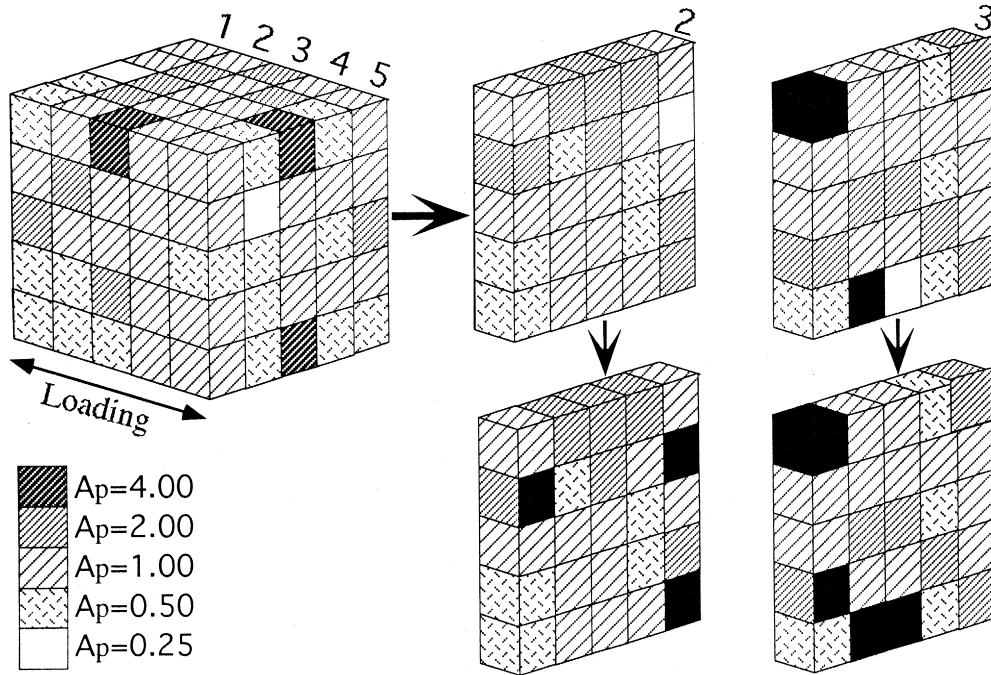


Fig. 10. The fracture process in NAV10 non-uniform model (the effect of particle aspect ratio).

strength is 630 MPa and the fracture strain is 0.06 in the SiC31% spherical model and they are 590 MPa and 0.025 for the SiC10% prolate model. This means if the SiC particle is rigid and perfectly bonded to matrix, it is better to keep the SiC particle aspect ratio equal to 1.0 and increase SiC particle volume fraction to improve the ultimate strength and fracture strain of the material.

#### 4.2. Fracture analysis of non-uniform model

**4.2.1. The non-uniform model.** In this study two types of non-uniform models are made. One considers the non-uniform distribution of SiC particle volume fraction and its aspect ratio is kept constant [Fig. 9(a)]. The other considers the non-uniform distribution of aspect ratio and keeps its SiC particle volume fraction constant (Fig. 10). Three non-uniform models are made for the former case. They are called RV02A1 and RV10A1 models when the aspect ratio 1.0 and the average particle volume fraction is 2 and 10%, respectively. The RV10A4 model means the aspect ratio is 4.0 and the average particle volume fraction is 10%. Two non-uniform models are made for the latter case. An RAV10 model is made using a random number to determine the particle aspect ratio's distribution. The other is called an NAV10 model in which the distribution of the particle aspect ratio is determined by the experimental result. There are only two arrangements of the aspect ratio; one is a long diameter parallel to the loading direction and the other is a short diameter parallel to the loading direction, half of each one is taken into consideration for the non-uniform model.

In the present study a box shaped super element [28] is used for a 3D non-uniform model as shown in Fig. 9(b). This element is built up of 24 constant strain tetrahedral elements. The mesh used for the non-uniform model is composed of  $5 \times 5 \times 5$  box super elements and each box represents 24 tetrahedral units as shown in Fig. 9(a). The location of each unit is determined by the random number. The average SiC volume fractions of the non-uniform models are 2 and 10%, respectively, which are the same as the SiC2% and SiC10% specimens used in the experiment. The average SiC particle aspect ratios are 1.0, 1.18, 1.18 and 4.0 for RV10A1, RAV10, NAV10 and RV10A4 models, respectively. By changing the random number system, two different non-uniform models for each RV02A1 and RV10A1 model are made. The boundary conditions are given that the normal displacement components along each side are assumed to be equal. The stress-strain relationships of each local area is decided by Figs 7 and 8. Each curve is approximated by the following equation:

$$\sigma = a_1 \varepsilon^4 + a_2 \varepsilon^3 + a_3 \varepsilon^2 + a_4 \varepsilon + a_5 \quad (3)$$

where  $a_1$  to  $a_5$  are constants and are determined by the best fit of the stress-strain curve. In each local unit, it is assumed that the fracture occurs if the equivalent strain or equivalent stress reaches the critical values of Figs 7 and 8 and the elements of fractured units are eliminated by releasing the nodal reaction forces corresponding to the remaining stresses in the element. The nodal forces in the element are reduced to zero in five stages.

**4.2.2. Results of the non-uniform model.** The result of the fracture analyses of the RV10A1 model is shown in Fig. 9(c). The black region means that the fracture occurs in this region. It is shown that the fracture first occurs in the second layer where the SiC particle local volume fraction is the largest (31% area) and does not yet occur in the third layer. Due to the increase of plastic deformation, the fracture areas spread on both sides and a new fracture occurs at different locations both in the second and the third layers, where the SiC particle local volume fraction is also large. Similar fracture processes are observed in the two different random system models for each of the RV02A1 and RV10A1 and RV10A4 models. The final fracture may occur by the linkage of these local fracture areas. It is consistent with the experimental observation that the particle volume fraction on the fracture surface is much larger than the average value.

The results of the NAV10 model are shown in Fig. 10. The second and the third layers from NAV10 model are also shown in Fig. 10. It is shown that the fracture occurs at first in the third layer where the reinforcement local aspect ratio is the largest ( $A_p = 4.0$  area). Due to the increase of plastic deformation, the fracture areas spread on both sides and a new fracture occurs at a different location, where reinforcement local aspect ratios are also the largest ( $A_p = 4.0$  area) or the smallest ( $A_p = 0.25$  area). The final fracture may occur by the linkage of these local fracture areas. The RAV10 model shows a similar fracture process. It is shown that the fracture more easily occurs in the oblate and prolate particle than in the spherical particle when the particle volume fraction is the same.

Figure 11 shows the stress-strain relationships of RV10A1, RAV10, NAV10 and RV10A4 non-uniform models. In Fig. 11, the uniform model (dashed line) and experimental results (circle symbol lines) are shown. It is shown that the ultimate stresses are nearly equal in all models except the RV10A4 model. The ultimate strength of RV10A4 is the largest of every model and the fracture strain of it is the smallest. This is because the average particle aspect ratio is the largest in every non-uniform

models. The interesting thing here is, though the average aspect ratio and particle volume fractions are nearly equal to each other in the RV10A1, RAV10, NAV10 and uniform models, the fracture strains changed from 0.03 to 0.15. The differences of fracture strain in these models are due to its non-uniform distribution of particle volume fraction and aspect ratios in the base matrix. In the uniform model, it is assumed that the particle aspect ratio is equal to 1.0 and the volume fractions are 10% in all local areas. So the fracture uniformly occurs in the matrix and the final fracture of the specimen is delayed. Then the fracture strain shows the largest value. In the case of the RV10A1 non-uniform model, it is assumed that the aspect ratio is equal to 1.0 in all units and the particle volume fractions change from 0 to 31%. Thus, the fracture occurs earlier in the area where the local particle volume fraction is the largest and the fracture strain is decreased largely (about 0.062) compared with the uniform models (about 0.15). In the RAV10 model, the fracture strain decreases about 0.026 and has a smaller fracture strain. In the last NAV10 model, the stress decreased suddenly when the strain reached 0.025. But after some amount of decreases it stopped and when the strain increased to 0.045 it dropped down again. It is shown that the fracture does not occur in an area where the particle shape is nearly spherical. These results imply that if the aspect ratios of all particles are close to 1.0 and the particles are distributed uniformly, the fracture strain increases a lot.

## 5. DISCUSSION

The present work has qualitatively identified the uniform and non-uniform damage process, as ductile failure in the local and global regions and obtained some reasonable damage phenomena through experimentation. However, the stress-strain relationships of each unit used in the non-uniform models are fitted from a uni-axial tension condition (uniform stress triaxiality condition as 0.33) and the unit models do not consider the effect of residual stress, particle mutual interaction, interface debonding and particle cracking.

The thermal residual stresses have been studied by Shen *et al.* [17]. The results show that the thermal residual stresses have a strong effect on the overall constitutive response during subsequent loading in tension or compression. Due to Shen *et al.* results, the effect of residual stress can be considered by changing the stress-strain relationships of the matrix slightly. If so the main results of the paper on the fracture process of the models do not change qualitatively. Moreover, it is extremely difficult to obtain the local residual stress field experimentally. For these reasons the analysis of the thermal residual stress is not considered in this study.

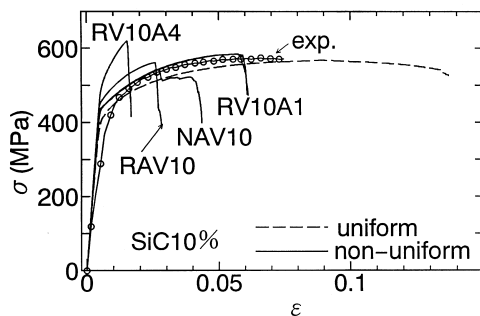


Fig. 11. Stress-strain relationship of SiC10% models.



The mutual interaction may change the local constraint and local stress triaxiality. The local damage pattern as local dimple fracture, interface debonding and particle cracking may vary under locally different constraint or stress triaxiality conditions. The non-uniform stress triaxiality or constraint effects may then affect the local and global damage process, but this problem is not considered in this work. The effects of stress triaxiality and constraint have been discussed by many authors [30, 31] using unit cell models by changing the stress triaxiality, with and without side constraint conditions. The results show that the microscopic fracture process is well simulated using a unit cell model qualitatively. However, the quantitative analysis of stress-strain relationships also have some difficulties. The reason for this is that the stress triaxiality and the side constraint change largely not only with the changes of the distance, location and geometrical shape of the particles but also during the deformation history. On the other hand, it is extremely difficult to obtain accurate results of the stress-strain relationship for changeable stress triaxiality and constraint areas for non-uniformly distributed particle matrix composites. Further study is necessary to overcome this difficulty and this will be pursued in the future.

## 6. CONCLUDING REMARKS

The effect of the non-uniformity of the SiC particle reinforced aluminum alloy on the dimple fracture process is studied using FEM analysis. The effect of non-uniform distribution of the particle and particle aspect ratio are evaluated using a non-uniform model. The following results are obtained.

(1) The SiC particle volume fraction and aspect ratio largely affect the stress-strain relationships and their effects increase by increasing the particle volume fraction. When the SiC particle is uniformly distributed in the base matrix, the tensile strength increases by increasing the particle aspect ratio. The fracture strain increases with the increase of the  $A_p$  value up to  $A_p = 1.0$  and it decreases again for  $A_p > 1.0$ .

(2) Through the analysis of a non-uniform model, it is shown that the local fracture occurs where the local particle volume fraction is the largest and/or the  $A_p$  value in local areas the maximum or minimum. The global fracture occurs by linkage of these locally fractured areas.

(3) According to the results of (1) and (2), it is pointed out that the mechanical properties of this composite material may be improved if the particle aspect ratios are close to 1.0 and the particles are distributed uniformly in the base matrix. In order to increase the ultimate strength and fracture strain of the material, it may be better to increase the particle volume fraction rather than increasing the particle aspect ratios if the SiC particle is rigid and perfectly bound to the matrix.

## REFERENCES

1. Lloyd, D. J., *Acta metall. mater.*, 1991, **39**(1), 59–71.
2. Lloyd, D. J., *Int. Mater. Rev.*, 1994, **39**(1), 1–21.
3. Geni, M., Kikuchi, M. and Hirano, K., *Trans. JSME A*, 1995, **61**(581), 45–51 (In Japanese).
4. Llorca, J., Needleman, A. and Suresh, S., *Acta metall. mater.*, 1991, **39**(10), 2317–2335.
5. Tohgo, K. and Weng, G. J., *Trans. ASME J. Engng Mater. Tech.*, 1994, **116**, 414–420.
6. Tao, S. and Boyd, J. D., *Proc. of ASM Mater. Congr.* 1993, 29–40.
7. Llorca, J., Suresh, S. and Needleman, A., *Metall. Trans. A*, 1992, **23**, 919–934.
8. Povirk, G. L., Stout, M. G., Bourke, M., Goldstone, J. A., Lawson, A. C., Lovato, M., Macewen, S. R., Nutt, S. R. and Needleman, A., *Acta metall. mater.*, 1992, **40**(9), 2391–2412.
9. Liaw, P. K., Shannon, R. E., Clark, W. G., Jr. and Harrigan, W. C., Jr., *Cyclic Deformation, Fracture and Non-destructive Evaluation of Advanced Materials*. ASTM, STP1157, 1992, pp. 251–277.
10. Morimoto, T., Yamaoka, T., Lilholt, H. and Taya, M., *Trans. ASME J. Engng Mater. Tech.*, 1988, **110**, 71–76.
11. Kamat, S. V., Hirth, J. P. and Mehrabian, R., *Acta metall. mater.*, 1989, **37**(9), 2395–2402.
12. Flom, Y. and Arsenault, R. J., *Acta metall. mater.*, 1989, **37**(9), 2413.
13. Kikuchi, M. and Geni, M., *Contemporary Research in Engineering Science*, ed. C. Batra Romesh. Springer, 1995, pp. 276–288.
14. Christman, T., Needleman, A. and Suresh, S., *Acta metall. mater.*, 1989, **37**(11), 3029–3050.
15. Kikuchi, M., Geni, M., Togo, T. and Hirano, K., *Mechanisms and Mechanics of Composite Fracture*, eds. R. B. Bhagat, S. G. Fishman and R. J. Arsenault. ASM International, 1993, pp. 97–106.
16. Finot, M., Shen, Y.-L., Needleman, A. and Suresh, S., *Metall. Trans. A*, 1994, **25**, 2403–2420.
17. Shen, Y.-L., Finot, M., Needleman, A. and Suresh, S., *Acta metall. mater.*, 1995, **43**(4), 1701–1722.
18. Nishiyama, K. et al., in *Proc. 4th Japan-U.S. Conf. on Composite Material*. Washington, DC, 1988, pp. 128–137.
19. *Encyclopedia of Advanced Material Applications*. SUNCHOH, 1990, p. 392 (in Japanese).
20. Lloyd, D. J., Lagace, H., McLeod, A. and Morris, P. L., *Mater. Sci. Engng A*, 1989, **107**, 73–80.
21. McClintock, F. A., *ASME J. appl. Mech.*, 1968, **35**, 363–371.
22. Rice, J. R. and Tracey, D. M., *J. Mech. Phys. Solids*, 1969, **17**, 201–217.
23. Gurson, A. L., *Trans. ASME J. Engng Mater. Technol.*, 1977, **99**, 2–15.
24. Tvergaard, V. and Needleman, A., *Acta metall. mater.*, 1984, **32**, 157–169.
25. Tvergaard, V., *Int. J. Fract.*, 1982, **18**(4), 237–252.
26. Okayasu, K., Miyamoto, H., Machida, K. and Kawazoe, T., *Trans. JSME*, 1987, **53**(488), 732–737 (in Japanese).
27. Brown, L. M. and Embury, J. D., *Proceedings of the 3rd International Conference on Strength of Metals and Alloys*. Inst. of Metals, London, 1973, pp. 164–169.
28. Andersson, H., *J. Mech. Phys. Solids*, 1977, **25**, 217–233.
29. Koplik, J. and Needleman, A., *Int. J. Solids Struct.*, 1988, **24**, 835–853.
30. Tvergaard, V., *Engng Fract. Mech.*, 1988, **31**(3), 421–436.
31. Kikuchi, M., Geni, M., Hirano, K., *Localised Damage IV, Computer Aided Assessment and Control*, eds. H. Nisitani, M. H. Aliabadi, S.-I. Nishida and D. J. Cartwright. Computational Mechanics Publications, 1996, pp. 423–432.